Mapping the South Pole Atmosphere with the Water Vapour Radiometer

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1 Summary

We have relooked at the WVR data and found some small systematic effects in timing, tilt and backlash which we can correct for.
We used an am multilayer model to fit PWV to every corrected 4-channel data point. These fits are good for elevations above 40 degrees and we showed that the PWV variation over time is much bigger than the fit error. From this position it will take us further few steps to produce Az maps of the atmosphere as a function of time.

2 Introduction

The final goal of this project and the main purpose of the WVR is understanding the South Pole Atmosphere to decouple atmospheric noise from real CMB Temperature fluctuations. With this goal in mind, we want to correlate WVR data with Bicep Pair sum and Pair diff data and see what fraction of the atmospheric noise remains in the data after pair differencing.
If the atmosphere is completely unpolarized, as it is supposed to be, and there is no systematic in the detectors, differencing the two detectors in one pair should completely remove the atmospheric noise. However having another independent way to monitor atmospheric fluctuations can help identifying systematics in the experiment, evaluating how well the atmospheric noise is removed after pair differencing and, in case it is not removed well enough, developing a new method for atmospheric noise subtraction.

The first milestone we want to achieve is understanding the instrument well enough to use each set of four radiometric temperatures it returns to extract one single Precipitable Water Vapour (PWV) value. A simple way to extract PWV from radiometric temperatures is described in Richard Hill's posting [1]. This recipe extracts an independent PWV value from each frequency channel, without using any spectroscopic information. We found that single PWV values extracted from different channels are not always in agreement. Following this method and using the 4 channels independently ignores the most important information that comes from having temperatures at 4 different frequencies: the shape of the water line.

A more robust way to extract the PWV, is to read all the four radiometric temperatures at once and use am to fit them with a single water line shape. Am takes an atmospheric profile, which includes informations on how pressure, temperature and PWV scale on different atmospheric layers and extracts a tropospheric PWV value (or, equivalently, N: the number density of H_2O).

We decided to temporarily leave aside the Az Scans and to focus on Elnods. The idea is to use Elnods - acquired every hour- to extract average hourly information on PWV and then use Az scans to detect small spatial fluctuations around the mean values. This is computationally quicker.

The extracted PWV information at 183 GHz - the center of the WVR observation window - can be used to extrapolate the atmospheric emission strength in the Bicep Array frequency windows.

Varying N, pressure (P), or temperature (T) separately give distinct spectral signatures across the WVR channels. In fact, from pressure broadening, thermal linewidth, etc. varying each of these three parameters has a different spectral effect. A plot illustrating each of the three cases is reported in Fig.1, using am and a single slab atmospheric model.

3 Reading out Radiometric Temperatures: Fast Files are the best option

Before extracting PWV values it is important to understand how the WVR works, how data are recorded and so how to collect radiometric temperatures

[1] http://bicep.rc.fas.harvard.edu/general_projects/analysis_logbook/20170711_wvr_pwv_estimate/datareduction.p

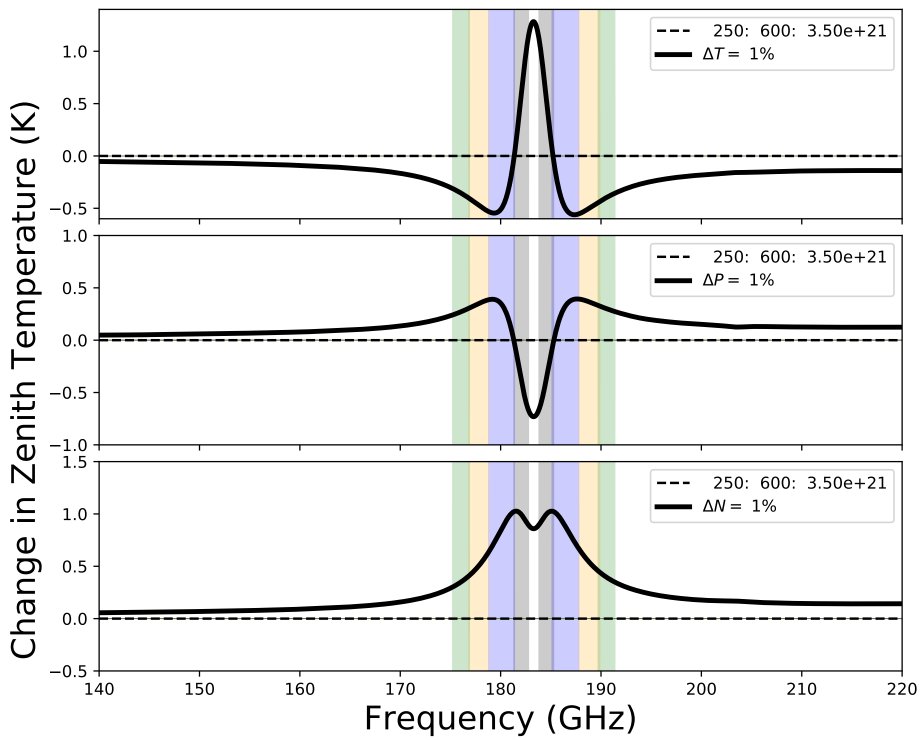


Figure 1: The spectral signature, in Kelvin, of 1% variations of T,P, and N from the spectral intensity when $(T;P;N) = (250K;600mm;3.5 \cdot 10^{21}/cm^2)$. The approx. bandwidths and frequencies of the four radiometer channels are shown in colours.
Top: A 1% change in the atmospheric temperature T results in a positive $T_{measured}$ derivative for Channel 0, and a negative derivative for the other 3 channels.
Center: A 1% change in the atmospheric pressure P results in a negative $T_{measured}$ derivative for Channel 0, and a positive derivative for the other 3 channels.
Bottom: A 1% change in the atmospheric water column density N results in a positive $T_{measured}$ derivative for all the four channels. This is the only scenario in which the responses of all four channels vary with the same sign.

from the radiometer data files optimally.
The WVR is a warm heterodyne radiometer. It has 4 Double-Sideband channels centered on the 183.31GHz water line and measures the sky temperature in those 4 frequency channels. It is equipped with an azimuth rotation stage that allows us to scan the beam in azimuth, and an elevation stepper motor to allow us to scan the beam in elevation.

The WVR records Elnod data in two different file formats which we call Fast Files and Slow Files. An internal mirror allows a set of successive measurements of Hot load, Position A, Cold Load, Position B, repeat. The internal mirror cadence is $0.056s$ and all four channels are read at once. A Fast File contains explicitly all these measurements. Position A and position B are V_{sky} measurements, while differences between Hot and Cold are used to calibrate the radiometer. In Slow Files, Fast File data are processed: filtered averages of the four positions are used to extract the gain, and from that the calibrated average of position A and B. So Fast Files contain raw and completely unprocessed voltage readings, sampled at $0.2s$, while Slow Files contain calibrated and smoothed Temperature data, sampled at $1s$. Slow Elevation Scan files are more straightforward to use, but - as a side effect - they contain artifacts due to data processing. They show a lag in angle of radiometer signals compared to elevation: this behaves as a single time-constant filter whose lag is consistent with a 7 degree backlash (see Fig.2).

To avoid adding systematics to the data we use in Fast Files to extract Radiometric Temperatures in the following analysis.
Working directly with fast files requires an intermediate step to interpolate and calibrate the data. The interpolation increases the fast files sampling rate of 4 times, and the calibration turns raw voltage readings into physical temperatures using the two T_{hot} and T_{cold} calibration sources.
For the calibration the following steps have been followed:

$$T_{ref} = \frac{T_{hot} + T_{cold}}{2} \tag{1}$$

$$V_{ref} = \frac{V_{hot} + V_{cold}}{2} \tag{2}$$

$$G = \frac{V_{hot} - V_{cold}}{T_{hot} - T_{cold}} \tag{3}$$

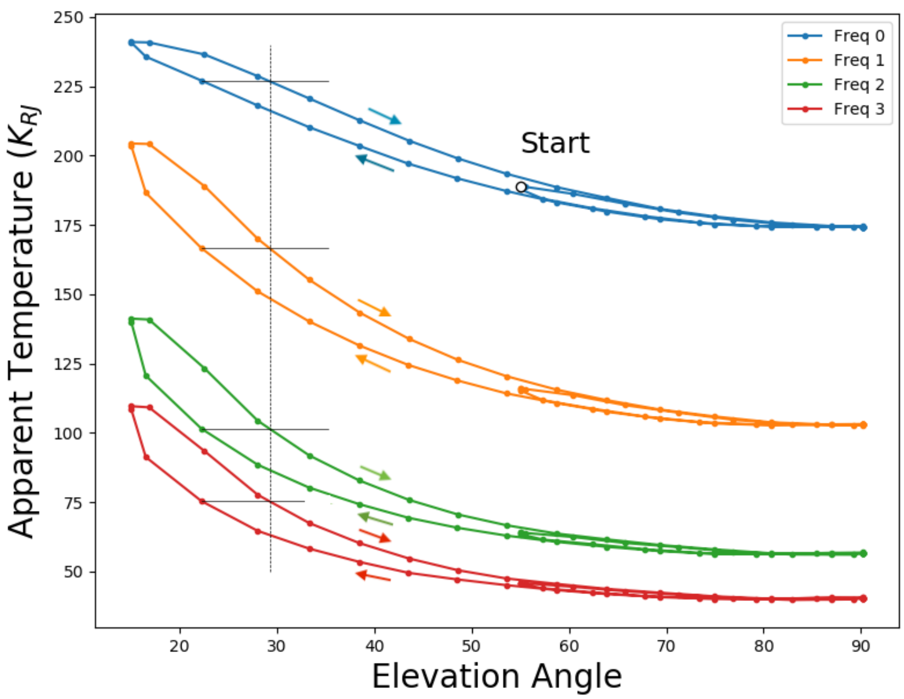


Figure 2: Radiometric Temperature vs elevation as read from a Slow File, for the four radiometer channels.
For all the four channels the T plots show a large hysteresis shape, consistent with a backlash of 7 deg (black lines in the figure). The hysteresis is evident especially at low elevations.

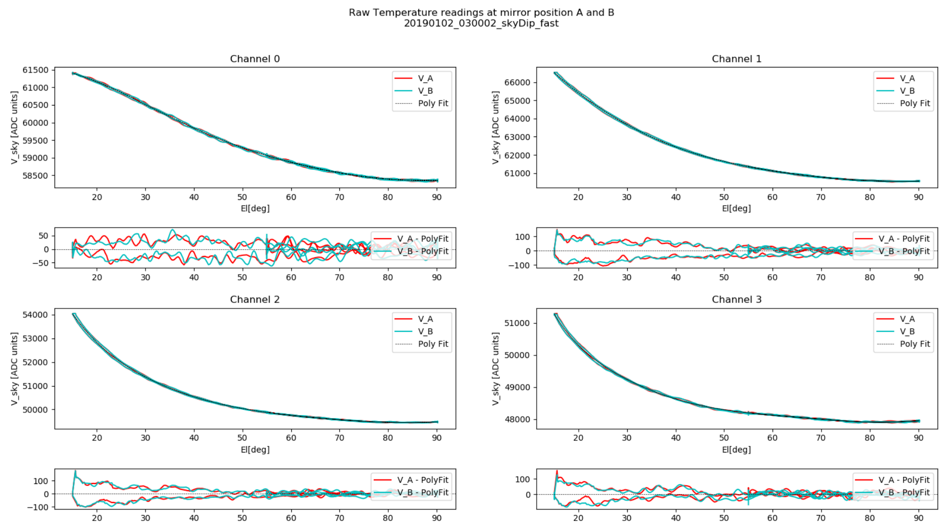


Figure 3: Raw temperatures readings at position A and B from a Fast file fitted with a third order polynomial. The plots shows the very good agreement between V_A and V_B readings.

$$T_{sky} = T_{ref} + \frac{V_{sky} - V_{ref}}{G} \tag{4}$$

Where T_{hot} and T_{cold} are the nominal temperatures of the two hot and cold reference sources, and V_{hot} and V_{cold} are the respective measured voltages. V_{sky} is the measured voltage when the radiometer is pointing at the sky. After interpolation, for every pointing position we have two WVR readings, one taken at mirror position A and another at Position B.

The Fast Files Elnods temperature data versus elevation are well fit by a third order polynomial.
Fitting fast-file radiometric temperatures to a polynomial in elevation angle we find the following things:

1. The WVR readings taken at mirror position A and at Position B agree with high precision.
   Therefore for all the analysis that follows we are going to use as V_{sky} at a given time/elevation the average of the two.

2. The variance is reduced to half inferring WVR gain from a whole file

instead of inferring an independent gain at each reading [Fig. 4].

3. Fast files are not lag free. They show a difference of 0.7 degrees between rising and falling data consistent with simple physical backlash [Fig. 5]. This is 10 times smaller than what we found for slow data .

4. Sometimes fast files show a zenith offset from nominal (EA=90). The offset value is generally small and varies quite a bit from elnod to elnod. This becomes evident plotting one month worth of effective zenith data [Fig. 7]. This point is still under investigation.

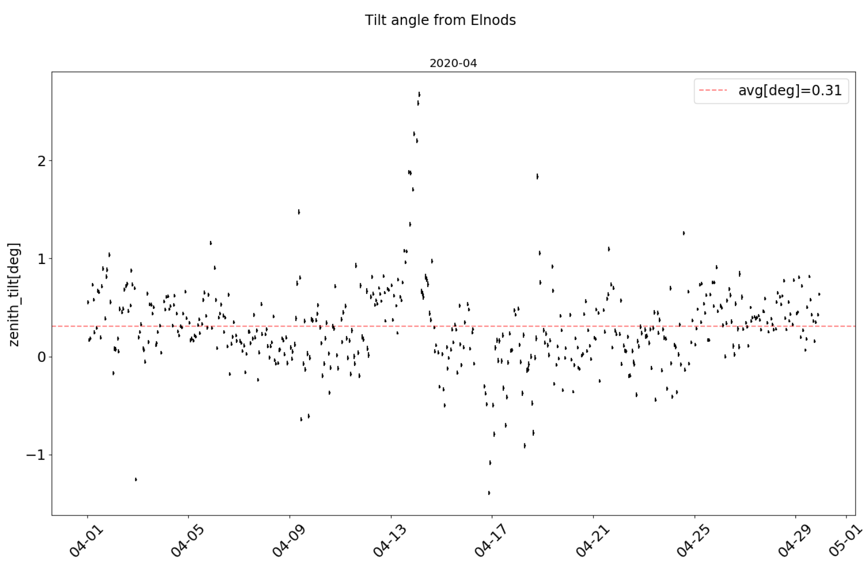
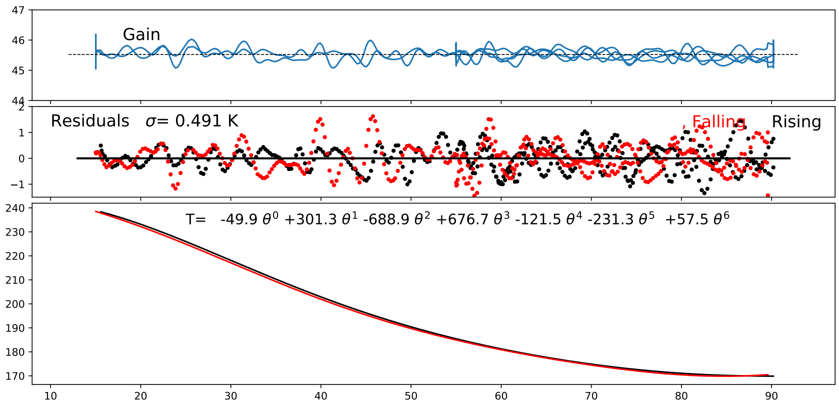
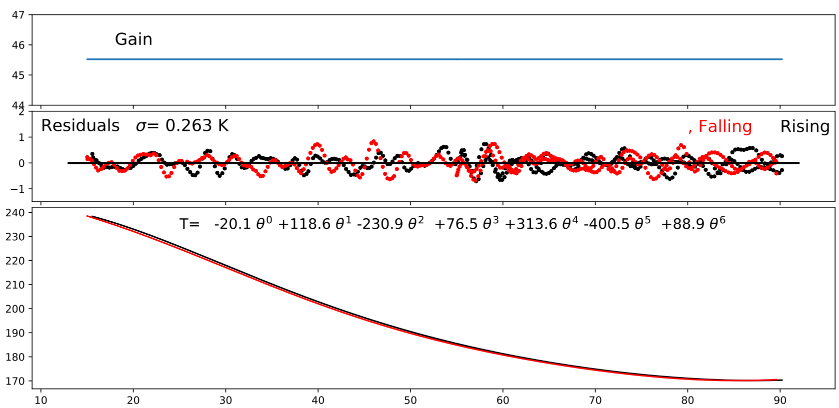


Figure 7: A scatter plot of the zenith tilt as a function of time for the month of April 2020. Each zenith offset value was extracted through a single slab atmospheric model fit: more details on that are reported in sect.4. The big drop in effective zenith angle at the center of the scatter plot is due to a set of elnods for which the single slab model doesn't properly describe the data, and therefore the returned value for the zenith offset is not accurate.

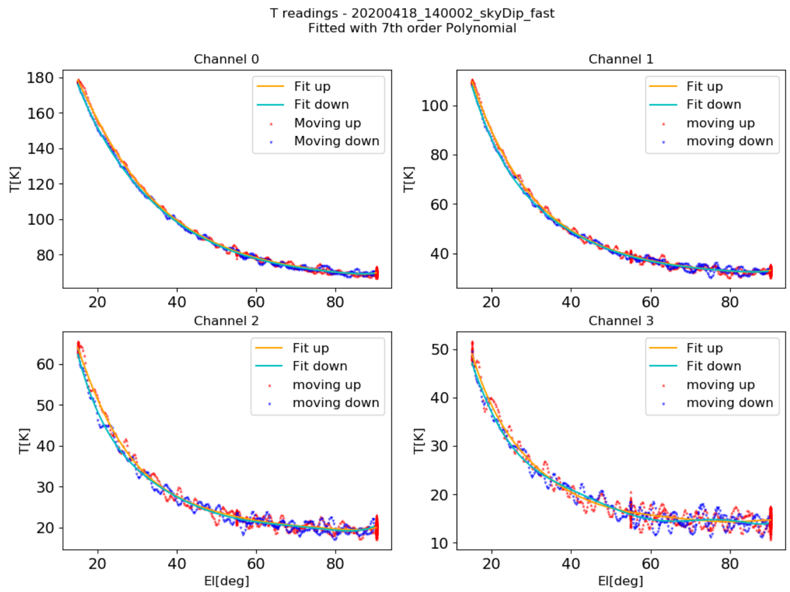


(a) Data calibrated using the local gain, point by point.

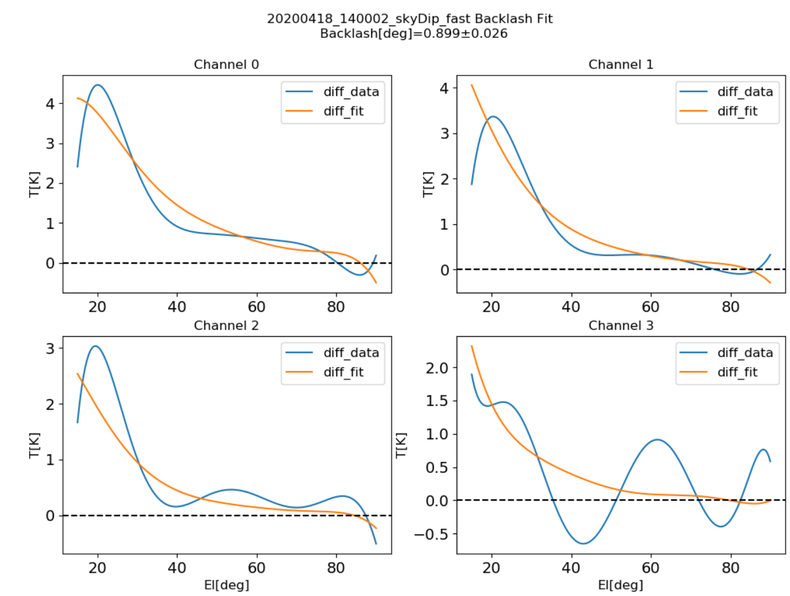


(b) Data calibrated using one single gain value, averaged over the full Elnod.

Figure 4: Two plots showing how the variance changes when data are calibrated using one single gain value, that is the gain average over the full Elnod, with respect to when the calibration is done using the local gain.



(a) Temperature Readings from a Fast File. The two different colors differentiate data acquired when the radiometer is moving up and when it is moving down.



(b) Difference between poly fits to rising and falling data, fitted with a simple backlash model.

Figure 5: The Fast Files temperature readings, showing a lag compatible with a simple backlash model.

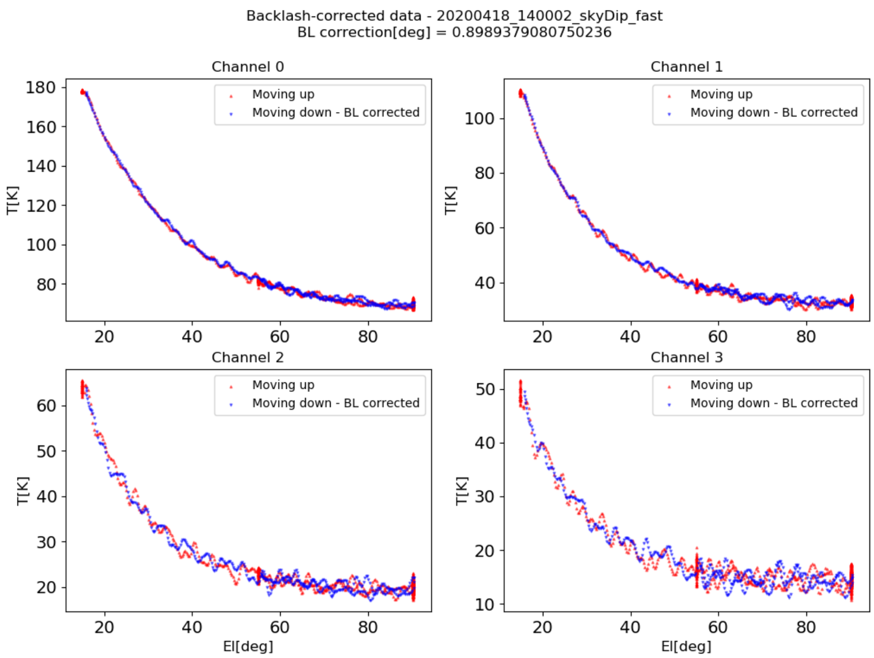


Figure 6: Temperature Readings from a Fast File after the backlash correction is applied.

4 A single slab model of the atmosphere is not enough to describe Elevation scans

The simplest assumption we can make is that the atmosphere is well described by a single homogeneous slab. This means that, for each radiometer channel, the measured T is can be written as:

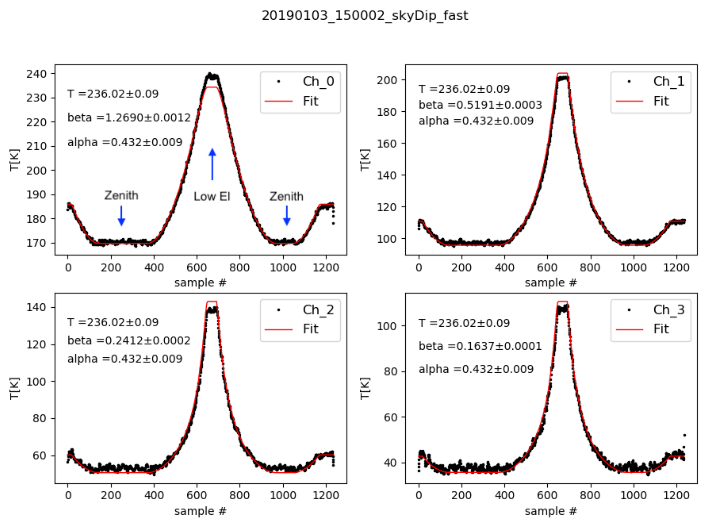
$$T_{sky} = T_{atm} \cdot (1 - \tau_{ch}) + 3K \cdot \tau_{ch} \tag{5}$$

Where the trasmission $\tau(\theta)$ is $\tau_{ch} = e^{-\frac{\beta_{ch}}{sin(\theta)}}$ and $\beta_{zenith} = -log(\tau)$.

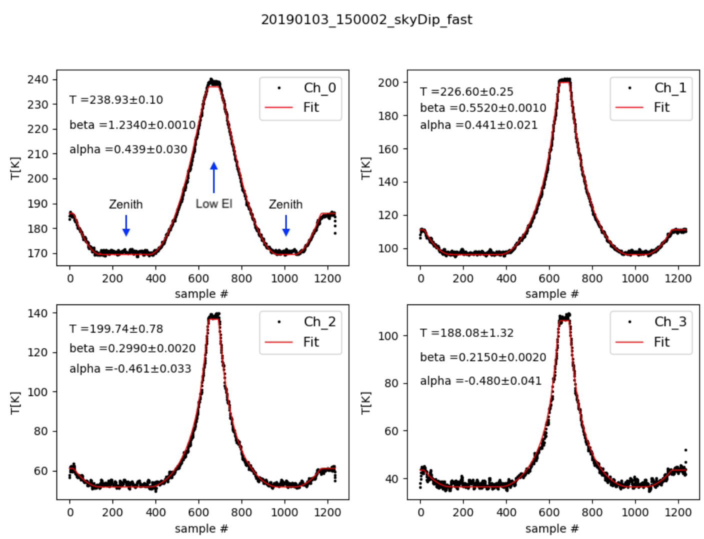
Given this model, the simplest thing we can do is fitting each of the four radiometer channels to a T$_{slab}$ and τ , where τ is free to vary with frequency (i.e. we have four different τ, one per channel) while T$_{slab}$ is constrained to be the same at all frequencies. For the data presented below we were still using data where the backlash shift was not corrected so we added to the model a time constant to account for the horizontal shift that we later identified as backlash (see Fig.5). So the final temperature of the sky is T_{sky} filtered including the averaging over a time constant α.
The fit to a single slab atmosphere doesn't turn out to be a very good fit. This fit works pretty well at medium elevations, but clearly doesn't work at low elevations and at zenith. If we try to leave T_{atm} also free to vary with frequency, the fit improves remarkably, but this doesn't have any physical sense. The result of the two kind of fits, with the two different kind of constraints, is reported in Fig.8. The poorness of the fit we obtained using the one layer model gave us motivation to move to a more complicated atmospheric model, later in this section.

Even if this simple fit is not accurate, it is not completely off. From the fit output - with T_{atm} constrained to be the same in all channels - we can take the four output opacities, and compare their frequency profile to an am line shape (see Fig.9). The zenith opacities found in the four channels are pretty well fit by a single uniform slab atmosphere with physical temperature $250K$, pressure $600mBar$ and column denstiy $3.5 \cdot 10^{21} molecules/cm^2$ calculated with am. This parameter combination corresponds to a precipitable water vapour of $1.047\mu m$.

The poor low elevation fits force us to consider a more complicated model. Am provides a 30 layer standard atmospheric model, in which each layer has a specific pressure and temperature and a specific dry air, H2O, and O3 column densities. And allows to fit for is the tropospheric Nscale, that is a



(a) The fit obtained constraining T_{atm} and α to be the same in all four channels. The fit is poor at zenith and especially at low elevation.



(b) The fit obtained leaving T_{atm} and α free to vary with frequency. This version of the fit is better but it doesn't have a simple physical meaning.

Figure 8: The Elnod profile against time fitted using the simple one layer atmospheric profile described by Eq.5. The three outputs parameter are the parameters from Eq.5 plus the time constant α described above. In this simple model a time constant $\alpha \sim 0.4$ succeeds in making the data symmetric. The poorness of this fit at zenith and low elevation, once we constrain T_{atm} to be the same in all four channels, forces us to consider a multi-layer atmospheric model.

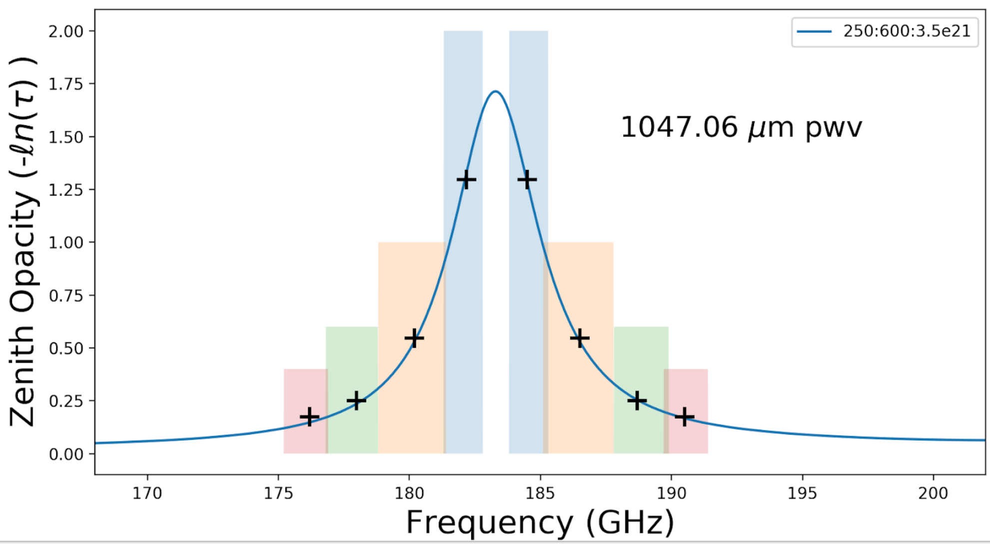


Figure 9: The four output opacities from Fig.8 to a single slab atmosphere with one T_{atm} and four τ_0. The blue curve is an AM line profile for a given Temperature, Pressure and PWV. This is not a proper fit but just a qualitative match. The line fits pretty well to the data points, even though the fits in Fig.8 are not perfect.

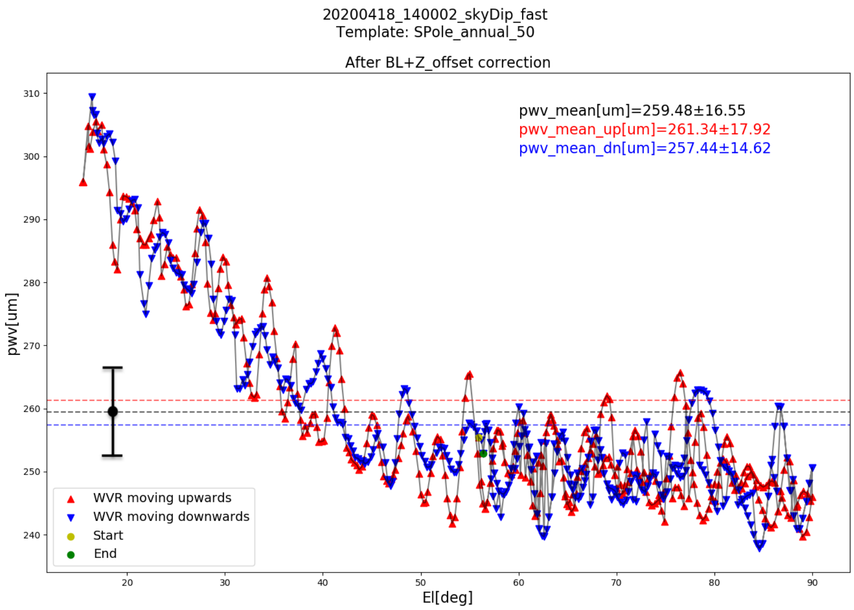


Figure 10: PWV values extracted from a full Elnod File, after Backlash correction. Each point is extracted performing an am fit and using a 30 layers atmosphere template. The plot shows that the 30-layer model fits the data well for elevation angles higher than $El \sim 50deg$, but it is not that good at lower elevations. Each red point in Fig. 11, corresponds to the average of the points in this kind of plot (i.e the black line), while the error bars correspond to the standard deviation. The data point extracted from this specific Elnod is reported on the left.

scale factor to the H2O column density in all the tropospheric layers.
Each Elnod data point was fitted independently using this template. The extracted PWV values are roughly constant in elevation for $El > 50deg$, but have a non-flat trend at lower elevation. As this deviation in PWV at low elevations tracks elevation angle, we concluded it is not due to atmospheric drift. This gives us a hint that also the 30 layers model is not completely correct.

Even with the large spread in PWV due to a poor model, PWV is inferred with high precision compared to variations over time. In fact, analyzing one month worth of elnod data we found that the monthly PWV std is $\sim 200\mu m$. While the PWV std over ~ 1 minute, which is the duration of one elnod, is

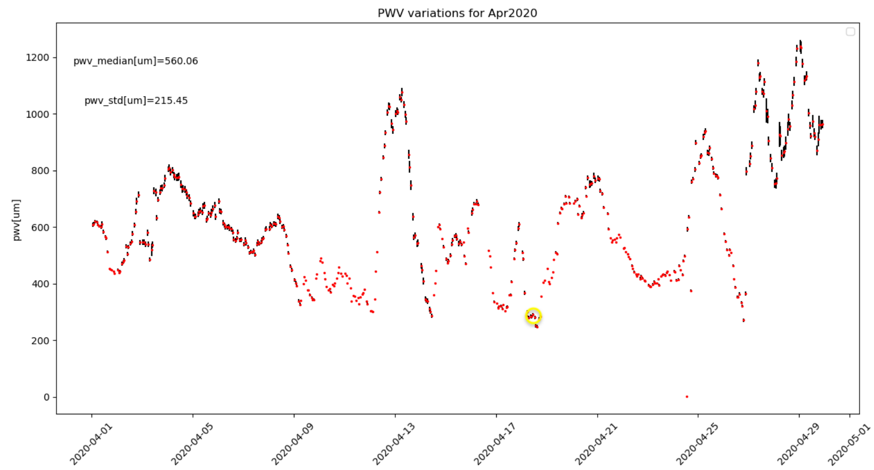


Figure 11: The plot reports the PWV variations measured for one month of data, in April 2020. Each data point is the average PWV value extracted from one specific Elnod in the reported day. The point extracted in Fig.10, is marked with a yellow circle. The daily and monthly variation in PWV appear to be of similar magnitude, and each is much greater than the variation on the time scale of a single measurement or the quality of the individual fits.

20 times smaller.

5 Zenith temperatures from the ElNod files tell us quite a bit about the atmosphere

Instead of reading out all elevations data, from one Elnod scan, we can read out just the Zenith temperatures. Monitoring the zenith temperatures for hours, weeks or months provides a lot of informations about how the atmosphere changes over shorter or longer time scales.
Important information can be extracted from comparing temperature fluctuations in different channels. Scatter plots of one WVR channel against another reveal that all the data lie along smooth curves. Fig 12 shows how data acquired in different days, even one year apart from each other, lie nicely along a smooth curve. These curves are well fitted by a line with a positive slope and a little bit of curvature, but are fully described by a second order

polynomial. Daily and hourly variations are smaller (10K and 1/2K) but they lie along the same curves.

The relative amplitudes of temperature fluctuations across channels indicate that this is mainly due to water vapour column density variations, with just small changes in atmospheric temperature and pressure (see Fig.1).
Moreover, analyzing one month worth of data it emerged that, in this time scale, zenith temperatures for channel 0 (the closest to the center of the line) vary by more than 100K. On the other hand, temperature records for February show ground thermal daily variations of $\sim +/-1K$. So the variation in T as measured by the WVR is one order of magnitude higher. This proves that measured variations in T, are not due to ground temperature variations.

6 Moving forward, we can use our understanding to invert the WVR files

Using the above information on instrument and instrument data, we can move forward and start analyzing Az scans to detect small PWV deviations from the mean value.
The interesting thing about Az scans is that they contain a systematic which actually gives us further information. In fact the radiometer is tilted, and therefore each AZ scan contains a dipole term which contains the same information as a full elevation scan. Once the dipole term is removed, temporal correlations between channels will allow an am fit to atmospheric parameters. From here, we are just a few steps away from having full PWV atmograms, that is maps of the Precipitable Water Vapour as a function of Azimuth and time.

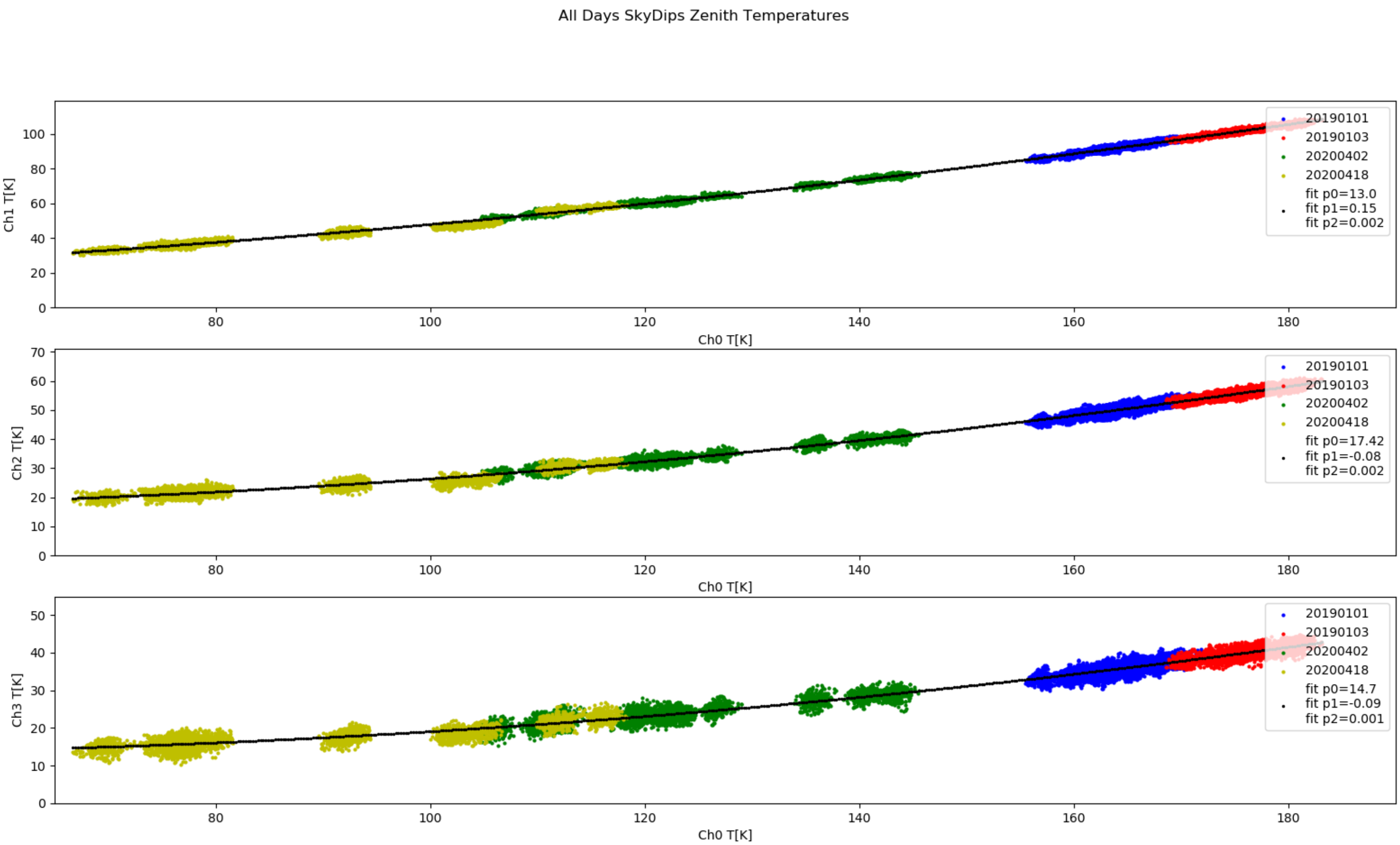


Figure 12: Zenith Temperature T_z for each channel plotted versus T_z for ch0, from four different days: 2019-01-01 , 2019-01-03, 2020-04-02, 2020-04-18. Even though the data set contains data acquired more than one year apart from each other, all of the data points lie on a single smooth curve with positive slope and a little curvature.